

## Context & pain point

### 400k MAU

EduNova Math & Science learning platform

### Students aged 11–16

middle school, mostly on mobile

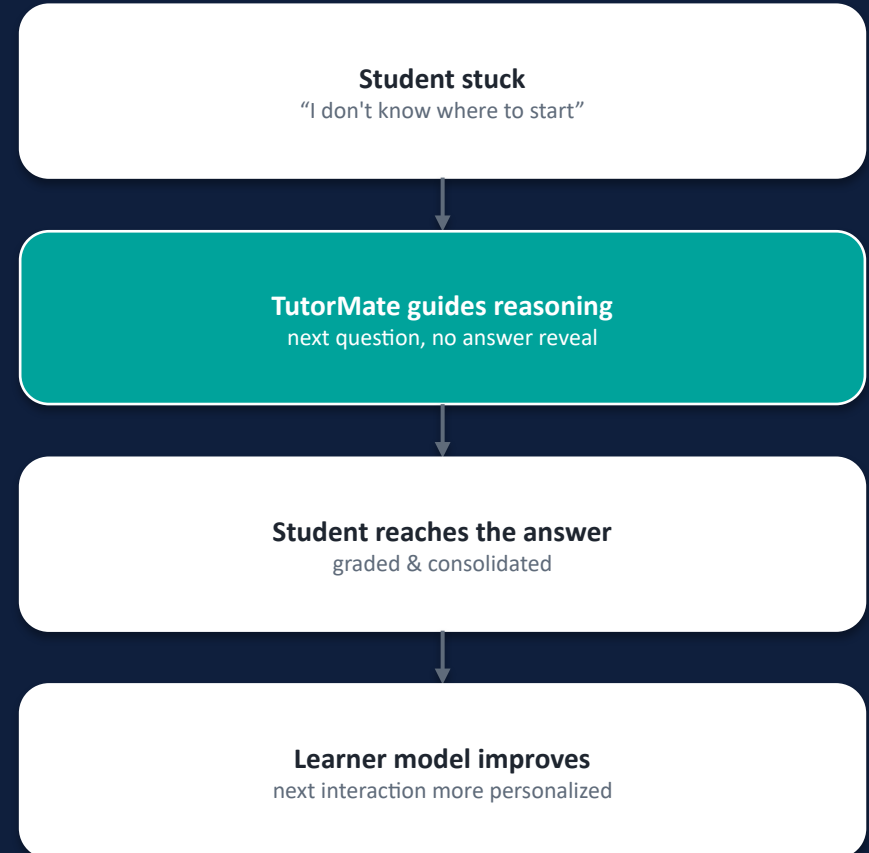
### Current gap

worked solutions do not teach reasoning — stuck → abandon

**TutorMate coaches students toward the answer instead of simply giving it.**

AI-First: remove the AI and there is no tutor left — only the old quiz app. All new value lives in the loop on the right.

## Value loop



# TutorMate ≠ ChatGPT with a prompt

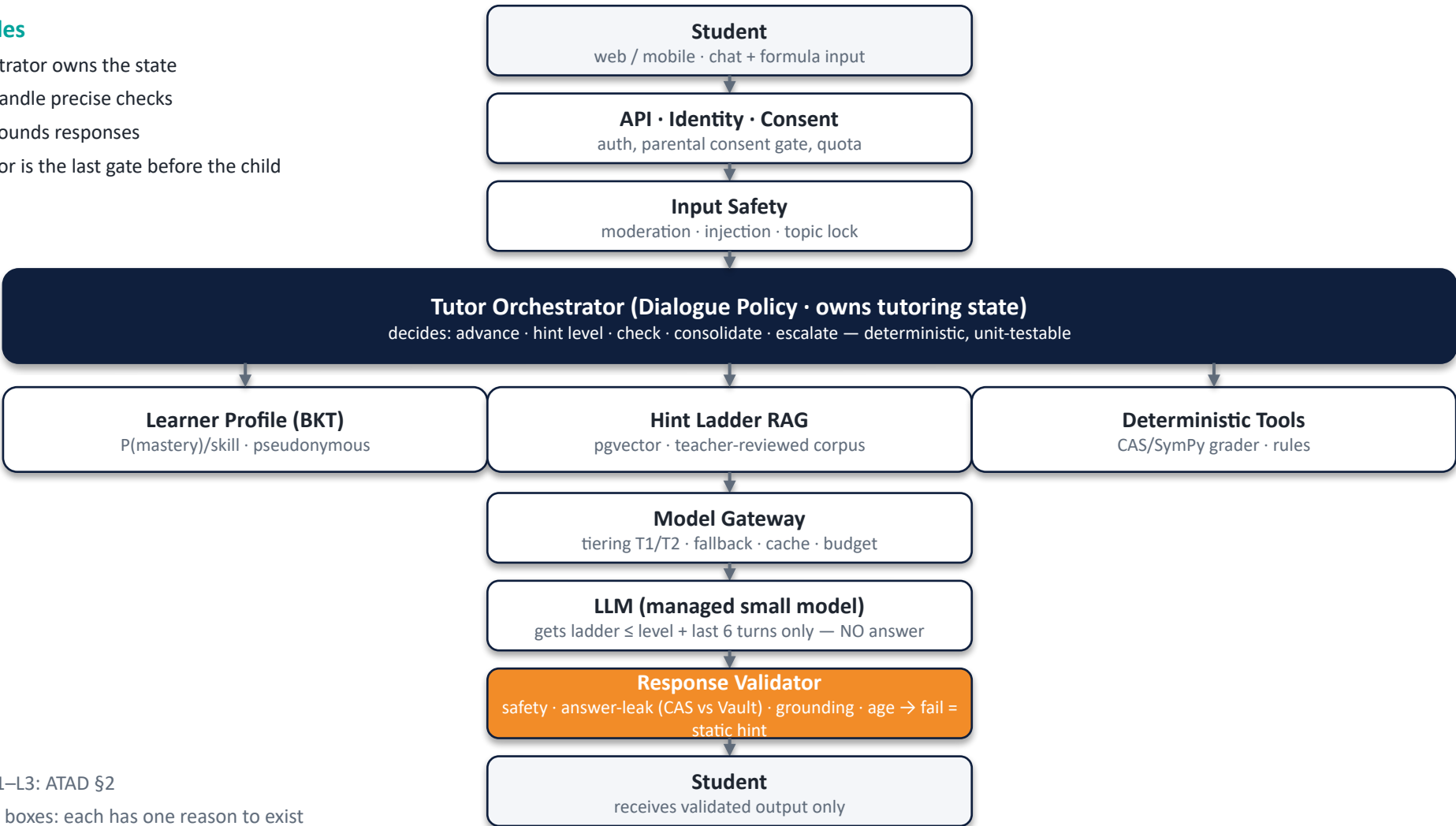
| General ChatGPT / generic chatbot | TutorMate                                              | Architectural mechanism                                       |
|-----------------------------------|--------------------------------------------------------|---------------------------------------------------------------|
| General-purpose                   | Bounded Math/Science tutor (domain-locked)             | Input Guard + topic classifier rejects off-scope              |
| Optimizes for ANSWERING           | Optimizes for LEARNING — next question, minimal reveal | Dialogue Policy picks the action; LLM only phrases it         |
| General model knowledge           | Approved curriculum grounding (Hint Ladder)            | RAG over teacher-reviewed, versioned corpus                   |
| Prompt-driven                     | Explicit tutoring state machine                        | Orient→Elicit→Hint→Check→Consolidate→Escalate (ADR-001)       |
| Output straight from model        | Output VALIDATED before it reaches the student         | Response Validator: safety · leak · grounding · age (ADR-004) |



# Target architecture: LLM ≠ policy engine

## Principles

- Orchestrator owns the state
- Tools handle precise checks
- RAG grounds responses
- Validator is the last gate before the child



## Async (Kafka)

- Telemetry / events
- Evaluation sampling (LLM-judge + teachers)
- Monitoring · alerting
- Cost tracking / turn
- Learner model update

## Platform

AWS · EKS (stateless, HPA/KEDA) · Aurora Postgres + pgvector · ElastiCache Redis · MSK Kafka · S3 · Bedrock/OpenAI via LiteLLM gateway

[Details: Appendix B](#)

Full C4 L1–L3: ATAD §2

Only ~10 boxes: each has one reason to exist

# Enforced pedagogy · Bounded personalization

Rubric 4.1 · 4.4

## Socratic state machine (Dialogue Policy)



- Hint level +1 ONLY after  $\geq 2$  distinct errors (graded by CAS) — a rule in code, not in the prompt
- Escalate (conditional reveal / hand-off to teacher) only when the ladder is exhausted or a safety flag fires
- LLM receives: action + ladder  $\leq$  current level + last 6 turns  $\rightarrow$  it only phrases

### Example

Student:  $2x + 5 = 15$  — “I don't know how to do this”

**x Chatbot:**  $x = 5$

**✓ TutorMate (Elicit):** “What could you do first to remove the +5 on the left side?”

$\rightarrow$  1st wrong try (“divide by 2”): Check  $\rightarrow$  misconception ‘order of operations’  $\rightarrow$  Hint L1, step not revealed

$\rightarrow$  Correct: Consolidate — “Why does subtracting 5 from both sides keep the equation balanced?” then mastery update

## Learner profile — what we store / don't store

| STORE (bounded learning signals)               | DO NOT STORE by default                                         |
|------------------------------------------------|-----------------------------------------------------------------|
| P(mastery) per skill — BKT, pseudonymous ID    | Full long-term transcript (redacted raw: 30 days, then deleted) |
| Most recent error category (misconception tag) | Raw child identity in the prompt (token only)                   |
| Hint history per skill                         | Psychological, emotional or non-learning behavioral profiling   |
| Confidence (optional self-report)              | Location, device, parent contact in the learner store           |
| profile_version + consent_version              | Data used for training without separate consent                 |

- Personalization = next skill/exercise, difficulty, starting hint level — derived from the mastery vector
- Student / parent can view & reset the profile; deletion on request  $\leq 30$  days (ADR-002, §5.7)

Personalization is based on bounded learning signals — not unrestricted memory.

# Why Hybrid AI · Curriculum grounding

Chosen pattern: Hybrid — policy + RAG + tools + LLM

| Capability                       | Technology pattern                          | Why not let the LLM do it                                         |
|----------------------------------|---------------------------------------------|-------------------------------------------------------------------|
| Natural-language tutoring        | LLM (managed small, T1)                     | — this is the genuinely linguistic part                           |
| Curriculum grounding             | RAG over teacher-reviewed Hint Ladder       | Model doesn't know EduNova's curriculum; needs versioning & audit |
| Correctness, formula equivalence | Deterministic tool (CAS/SymPy)              | LLM arithmetic is unreliable; zero token cost                     |
| Mastery update                   | Deterministic logic (BKT)                   | Must be explainable, stable, auditable                            |
| Pedagogical control              | State machine (Dialogue Policy)             | Prompts can't enforce; code is unit-testable                      |
| Safety / answer leakage          | Validator (rules + classifier + safety API) | Never let the model grade itself                                  |

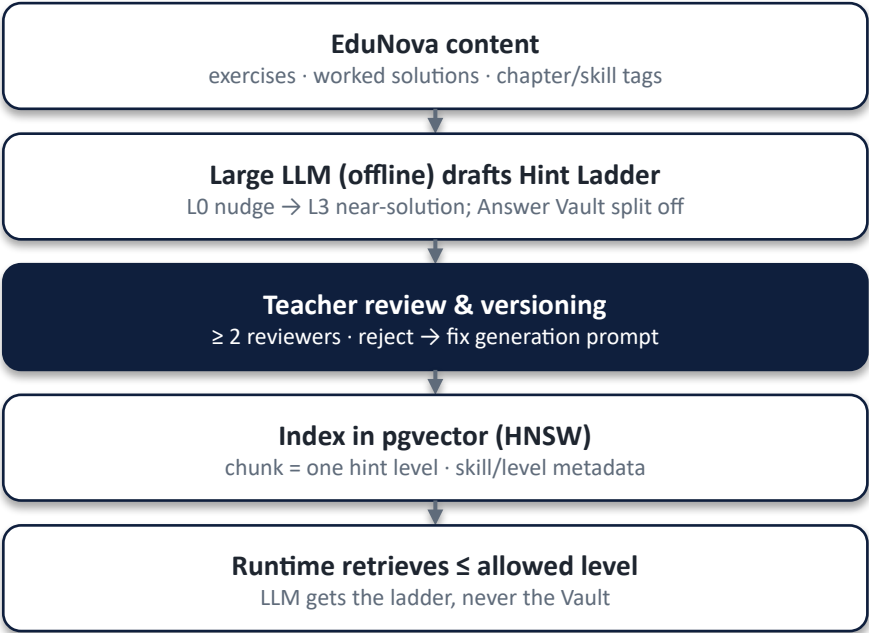
Why not fine-tune first?

No trusted tutoring dialogue corpus (~0) + curriculum changes frequently → RAG updates in place, fine-tuning must retrain. SFT/DPO of an 8B model is Phase 2, once ≥ 50k labelled turns exist (ADR-005).

Why not autonomous agents?

Unnecessary autonomy — the tutoring flow is finite; agents add latency, cost and safety surface.

## Curriculum grounding — offline lane



Use an LLM only where the problem is genuinely linguistic; grounding = an approved corpus, not the model's memory.

# Safety for minors: four layers of defence

1 • BEFORE THE MODEL

Identity · parental consent gate · input moderation · prompt-injection detection · topic lock

2 • DURING TUTORING

Bounded tools · approved retrieval only · state policy · token/turn budget · no answer in context

3 • BEFORE DELIVERY

Validator: safety · grounding · age-appropriateness · answer-leak (CAS vs Vault + paraphrase clf) → fail = static hint

4 • AFTER DELIVERY

Monitoring · continuous red-team · canary/rollback 5 min · kill switch → Static Ladder Mode

UNVALIDATED MODEL OUTPUT NEVER REACHES THE STUDENT

The two most important risks

| Risk                                                              | Controls                                                                                                        | Target                                    |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Harmful content (self-harm, violence, adult, grooming)            | Layers 1 + 3: classifier + managed safety API; self-harm → protocol: stop tutoring, show hotline, notify parent | 0 red-team pass-through · < 0.01% prod    |
| Answer leakage (“just give me the answer”, jailbreak, paraphrase) | State machine + canonical answer check (CAS) + leak classifier + adversarial eval every release                 | ALR ≤ 1.0% golden set · ≤ 0.5% production |

- Data minimisation: pseudonymous, redacted raw 30 days, no training without consent
- Bias/fairness: ALR & pedagogy score sliced by gender, region, free/paid
- XAI: every hint has a trace → ladder version · policy state · validator verdict

Governance mapped to NIST AI RMF · ISO/IEC 42001 · Vietnam Decree 13/2023 (PDP) · COPPA/GDPR-K if expanding — jurisdiction-specific compliance remains an ASSUMPTION pending legal confirmation (ATAD §11).

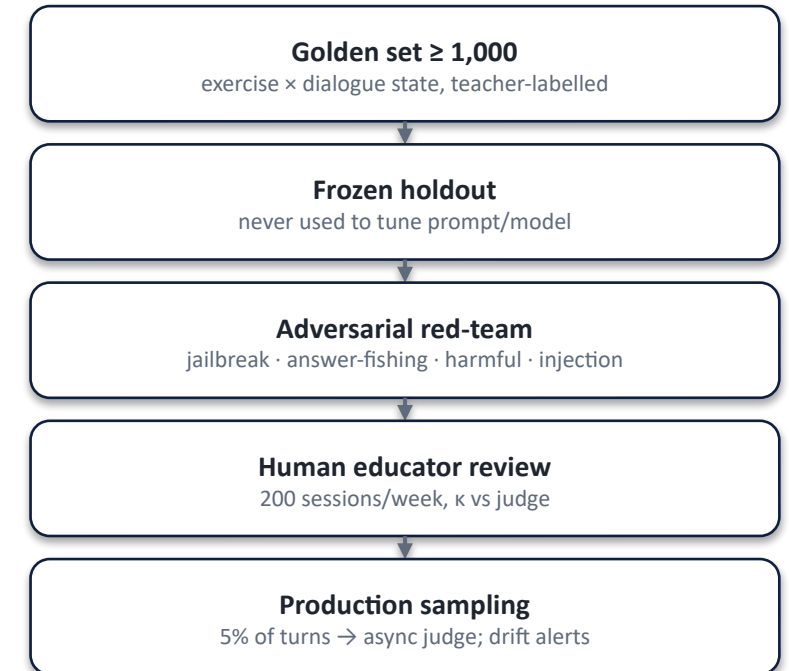
# Evaluation: how we know TutorMate teaches well

| Dimension   | Metric                                             | How measured                                           | Release threshold           |
|-------------|----------------------------------------------------|--------------------------------------------------------|-----------------------------|
| Correctness | Answer / step correctness                          | CAS + golden set                                       | ≥ 98% grader accuracy       |
| Grounding   | Hint has a valid source in the ladder              | Citation check                                         | ≥ 97%                       |
| Pedagogy    | State adherence · hint usefulness (Pedagogy Score) | LLM-judge calibrated vs teachers ( $\kappa \geq 0.7$ ) | ≥ 4.0/5                     |
| Leakage     | Answer Leak Rate (ALR)                             | CAS vs Vault + clf, adversarial set                    | ≤ 1.0% golden · ≤ 0.5% prod |
| Safety      | Critical escape · false refusal                    | Red-team suite 300+ cases                              | 0 escapes · FR < 3%         |
| Adaptation  | Mastery update in the right direction              | Offline BKT replay                                     | AUC ≥ 0.75                  |
| Performance | Full-turn latency                                  | Load test + prod                                       | p95 < 3s                    |
| Cost        | Blended cost / turn                                | Gateway ledger                                         | ≤ \$0.0007                  |

**RELEASE GATE:** quality + safety + latency + cost — ALL FOUR must pass before promotion (shadow → 5% → 25% → 50% → 100%)

- Real outcomes (not proxies): abandonment 35% → ≤ 22% · next-day same-skill accuracy 48% → ≥ 60% · Free→Paid 5% → ≥ 7% — measured by A/B in prod

## Datasets & process



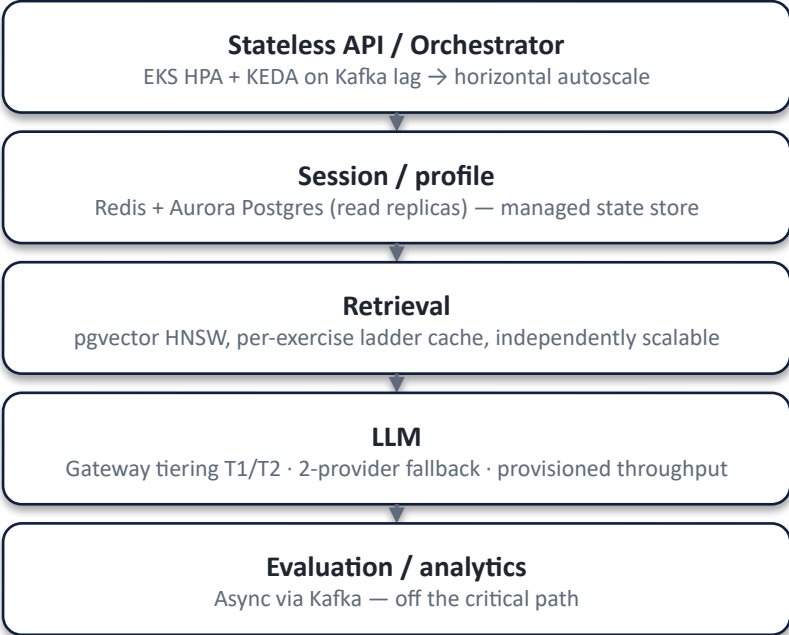
# Performance & Scale: p95 < 3s and 5–8× peak

## Load (turns/s)

| Level                   | Turns/s |
|-------------------------|---------|
| Average (960k/day)      | ≈ 11    |
| 5× peak                 | ≈ 56    |
| 8× peak (design target) | ≈ 89    |
| 10× load test           | ≈ 111   |

[A] 30% of DAU use the tutor × 8 turns; re-measure in 2k-student beta.  
Peak = 19–22h on exam days.

## Scaling strategy



## Latency budget (p95 < 3s)

| Step                                       | Budget     |
|--------------------------------------------|------------|
| Policy · profile · retrieval · input guard | < 0.7s     |
| LLM generation (T1, ~120 tokens)           | 1.5–1.8s   |
| Validation + mobile network                | < 0.5s     |
| Headroom                                   | ≈ 0.2–0.4s |

- First token streamed < 1s for perceived speed
- 12% of turns skip the LLM (CAS correct → static hint)
- Staging model p95 ≈ 2.6s — an architecture BUDGET, not a measured benchmark

**First expected bottleneck: model-provider concurrency / latency — not the stateless API tier → tiering + fallback + provisioned throughput.**

- Graceful degradation: provider failure → T2 on provider 2 → Static Ladder Mode (tutor SLO 99.5% incl. degraded; LLM-mode 99.0%)
- Vendor-neutral: Kubernetes · Postgres · Kafka · S3 API · LiteLLM — switching cloud/provider does not change the application

# Cost model & production operability

## Cost equation

**Monthly AI cost = turn volume × cost/turn + shared infra**  
 $\approx 29\text{M turns} \times \$0.00069 \approx \$18.4\text{k/month AI} \cdot + \text{infra} \approx \$11\text{k} \cdot \text{budget} = 8,000 \times P \rightarrow P \geq \sim \$2.30$

| # | Cost driver                            | Control                                                                         |
|---|----------------------------------------|---------------------------------------------------------------------------------|
| 1 | LLM tokens (prompt + history + output) | Model tiering T1/T2 · context limits · prompt cache · 12% of turns skip the LLM |
| 2 | Moderation / safety API                | Rules + small classifiers first; API only as backstop                           |
| 3 | Retrieval / embeddings                 | Embed offline once per ladder version · per-exercise cache                      |
| 4 | Retries / fallback / judge             | Turn budget · 5% judge sampling · quota-capped T2 fallback                      |
| 5 | Infrastructure                         | Spot for batch · scale-to-zero off-hours · managed services                     |

Free tier: 20 turns/day quota; paid unlimited · cost alert at 80% / kill at 120% of daily budget · [A] provider prices & 10% paying mix (sens. 5/15%) re-checked weekly

## Production operations

**Model Gateway fallback**  
T1 → T2 → provider 2 → Static Ladder Mode

**Release manifest**  
model + prompt + ladder + policy version, signed & stored

**Canary rollout**  
shadow → 5% → 25% → 50% → 100%, 4-condition gate

**Rollback ≤ 5 min**  
flip alias / prompt pointer, no redeploy

**Kill switch**  
disable LLM-mode globally or per cohort

**Cost & drift alerts**  
ledger / turn · ALR · daily judge score

- MLOps/LLMOps: eval CI on the golden set for every prompt/ladder PR · versioned registry · audit log
- On-call runbooks: leak spike, provider outage, harmful escape, cost spike

**Production readiness = the system fails safely, degrades gracefully, and stays within budget.**

# Five architecture decisions to remember

1

**Hybrid AI, not prompt-only**

LLM only generates language; RAG + CAS + BKT + state machine do the rest (ADR-003, -005)

2

**Explicit Socratic state machine**

Policy decides what is revealed; hint level is code, testable (ADR-001)

3

**Minimal learner profile**

Pseudonymous P(mastery)/skill; no long-term transcript, no PII in the prompt (ADR-002)

4

**Safety + validation BEFORE delivery**

Four defence layers; LLM never sees the answer; unvalidated output never reaches the child (ADR-004)

5

**Eval-gated, scalable, cost-controlled**

4-condition release gate · stateless autoscale · tiering + quotas  $\leq \$0.0007/\text{turn}$

**TutorMate is therefore not just an LLM chatbot. It is a controlled AI tutoring platform designed to be safe for children, measurable, and operable at scale.**

Team ownership: Thuận (Req & Integration) · Huyền (Logical & Data) · Hòa (AI/ML) · Phụng (Responsible AI) · Bình (Cloud & Scale) · Phú (Eval & Cost) · Dũng (Slides & Q&A) — Q&A moderated by Dũng; detailed allocation: Appendix D & ATAD

# Appendix A · Technology Selection Matrix

| Pattern                            | Fit               | Cold-start (~0 dialogue)   | Answer-leak control      | Latency/Cost     | Decision             |
|------------------------------------|-------------------|----------------------------|--------------------------|------------------|----------------------|
| Pure RAG + prompt                  | Medium            | Good                       | Weak (prompt-only)       | Good             | ✗ cannot enforce     |
| Fine-tune first                    | High (given data) | No data                    | Medium                   | Good (self-host) | ✗ Phase 2 (D-04)     |
| Autonomous agent                   | Low               | Good                       | Weak, hard to audit      | Poor             | ✗                    |
| Hybrid: policy + RAG + tools + LLM | High              | Good (synthesised ladders) | Strong (least-privilege) | Good             | ✓ Accepted (ADR-003) |

| Layer            | Candidates                                                     | MVP choice                                         | Rationale                                                            |
|------------------|----------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------|
| Runtime LLM (T1) | GPT-4o-mini · Claude Haiku · Gemini Flash · Llama 8B self-host | Managed small model via LiteLLM, 2 providers       | Latency, cost, no early GPU ops; self-host once ≥ 50k labelled turns |
| Offline LLM (T3) | GPT-4o · Claude Sonnet · Gemini Pro                            | Frontier for ladder generation / judge / synthetic | Quality matters more than cost; runs in batch                        |
| Vector store     | pgvector · Pinecone · Weaviate · OpenSearch                    | Postgres + pgvector (HNSW)                         | Small corpus (< 5M chunks), one system, ACID with metadata           |
| Grader           | LLM · SymPy/CAS · Wolfram                                      | SymPy + rules                                      | Deterministic, free, explainable                                     |
| Learner model    | BKT · DKT (deep) · IRT                                         | BKT                                                | Little data, explainable; DKT once enough events                     |
| Cloud            | AWS · GCP · Azure                                              | AWS (EKS, Aurora, MSK, Bedrock)                    | Team experience; portable via K8s/Postgres/Kafka/S3 API              |
| Gateway          | LiteLLM · Portkey · in-house                                   | LiteLLM                                            | OSS, tiering, fallback, cache, cost ledger                           |

# Appendix B · Cloud deployment (AWS, neutral)

| Layer           | AWS component                                                 | MVP → 8× peak configuration                                   | Neutral alternative           |
|-----------------|---------------------------------------------------------------|---------------------------------------------------------------|-------------------------------|
| Region / HA     | ap-southeast-1, 3 AZ                                          | Multi-AZ; warm DR in ap-northeast-1 (RPO 15 min)              | Any 3-AZ region               |
| Network         | VPC private subnets · NAT · PrivateLink to Bedrock            | Egress only via gateway allowlist                             | Equivalent on any cloud       |
| Edge / API      | CloudFront · WAF · ALB · API Gateway (WebSocket streaming)    | Rate-limit / quota per user & cohort                          | Cloud LB + WAF                |
| Compute         | EKS (Graviton) · HPA + KEDA (Kafka lag)                       | Orchestrator 6 → 48 pods · Guard 4 → 32 · Grader 4 → 24       | Any Kubernetes                |
| State           | Aurora PostgreSQL + pgvector · read replicas                  | r7g.xlarge ×2 → ×4; ladder cache in Redis                     | Any Postgres                  |
| Cache / session | ElastiCache Redis cluster                                     | 6 shards · 2h session TTL                                     | Redis / Valkey                |
| Streaming       | MSK (Kafka)                                                   | 3 brokers · topics events/eval/cost                           | Kafka / Redpanda              |
| Object storage  | S3 (ladder versions, redacted transcripts 30d, eval sets)     | Lifecycle 30d → glacier / delete                              | Any S3 API                    |
| LLM             | Bedrock (T1/T2) + OpenAI (fallback) via LiteLLM               | Provisioned throughput for peak; per-tier quotas              | Provider swap = config change |
| Observability   | CloudWatch · OpenTelemetry · Grafana · Langfuse-style tracing | Trace per turn: policy state · ladder ver · validator verdict | OTel-native                   |
| Security        | KMS · Secrets Manager · IAM least-privilege · GuardDuty       | Vault readable only by Grader/Validator roles                 | Equivalent                    |

GPU strategy: MVP owns no GPUs (managed inference). Phase 2: vLLM on g6/L4 or Inferentia for the 8B SFT model once data allows — decided on measured cost/turn (ADR-005). Details: ATAD §2.3, §10.

# Appendix C · Risk Register (top 8)

| ID   | Risk                                                   | L | I      | Key controls                                                                        | Owner |
|------|--------------------------------------------------------|---|--------|-------------------------------------------------------------------------------------|-------|
| R-01 | Answer leakage via jailbreak / paraphrase              | M | H      | LLM never sees the Vault · CAS-leak check · leak clf · adversarial eval per release | Phụng |
| R-02 | Harmful content reaching a child (self-harm, grooming) | L | Very H | 3-layer guard · self-harm protocol · 0 red-team escape gate                         | Phụng |
| R-03 | Factually wrong / off-curriculum hint                  | M | H      | Teacher-reviewed ladder · CAS · grounding check ≥ 97%                               | Hòa   |
| R-04 | Child-data breach (consent, retention)                 | L | Very H | Consent gate · pseudonymous · 30d retention · DPIA                                  | Phụng |
| R-05 | LLM provider outage / latency spike                    | M | M      | 2 providers · provisioned throughput · Static Ladder Mode                           | Bình  |
| R-06 | 8× peak exceeds forecast                               | M | M      | HPA/KEDA · free-tier quota · 10× load test                                          | Bình  |
| R-07 | Cost exceeds 20%-of-revenue ceiling                    | M | M      | Tiering · cache · quotas · alert 80% / kill 120%                                    | Phú   |
| R-08 | Cold-start: weak synthesised ladders, low adoption     | M | H      | 2k-student beta · teacher review · weekly abandonment KPI                           | Hòa   |

L = likelihood · I = impact. Reviewed every 2 weeks; R-01/R-02 have daily KRIs (ALR, escape count). Full 15-item register + prompt-injection threat model: ATAD §8, §11.5.

# Appendix D · Requirement · Assumption · Proposal

[R] Requirement = given by the workshop brief, non-negotiable · [A] Working assumption = ours, to be validated in beta · [P] Architect proposal = ours, backed by an ADR · [D] Data/training prerequisite = condition to enable a component

| Type             | Statement                                                                       | Source / validation                                   | Owner |
|------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|-------|
| [R] Requirement  | Full-turn p95 latency < 3s                                                      | Scenario B brief                                      | Bình  |
| [R] Requirement  | Sustain 5–8× peak; no direct answer reveal; parental consent; freemium          | Scenario B brief                                      | Thuận |
| [A] Assumption   | 30% of DAU use the tutor × 8 turns → 960k turns/day, 29M/month                  | Measure in 2k-student beta; refresh cost model weekly | Phú   |
| [A] Assumption   | Vietnam-primary market (Decree 13); 10% paying mix (sens. 5/15%)                | Legal + finance confirmation                          | Phụng |
| [A] Assumption   | Managed small model is good enough for Socratic phrasing given ladder + policy  | Golden set ≥ 1,000, Pedagogy ≥ 4.0/5 in beta          | Hòa   |
| [P] Proposal     | ALR ≤ 1.0% golden · ≤ 0.5% prod · golden set ≥ 1,000 · 4-condition release gate | ADR-001/-004; panel may challenge thresholds          | Phụng |
| [P] Proposal     | Hybrid policy + RAG + tools + managed LLM; AWS + pgvector; BKT                  | ADR-001....-005 (Accepted)                            | Hòa   |
| [P] Proposal     | Cost ≤ \$0.0007/turn; AI ≤ 20% of revenue                                       | Cost model §10; measured via gateway ledger           | Phú   |
| [D] Prerequisite | ≥ 50k real labelled turns before SFT/DPO of a self-hosted 8B (Phase 2)          | ADR-005 Proposed — not yet enabled                    | Hòa   |
| [D] Prerequisite | Teacher-reviewed ladders for ≥ 80% of top-traffic exercises before launch       | Offline lane, review tracker                          | Huyền |

Full 25+ item register: ATAD Appendix E. Every ADR carries a Status (Accepted / Proposed) and activation condition — nothing is presented as 'measured in production'.